
```
%sheikh wasimuddin lab 9 CSEIOT61
clc;
clear;
close all;

nodes=6;
steps=30;
nodeState=ones(nodes, steps);
eventData=floor(rand(nodes, steps)*4)+1;

for n=1:nodes
    currentState=1;
    for k=1:steps
        switch currentState
            case 1
                if eventData(n,k)==1
                    currentState=2;
                end
            case 2
                if eventData(n,k)==2
                    currentState=3;
                elseif eventData(n,k)==4
                    currentState=1;
                end
            case 3
                if eventData(n,k)==3
                    currentState=4;
                end
            case 4
                if eventData(n,k)==4
                    currentState=1;
                end
        end
        nodeState(n,k)=currentState;
    end
end

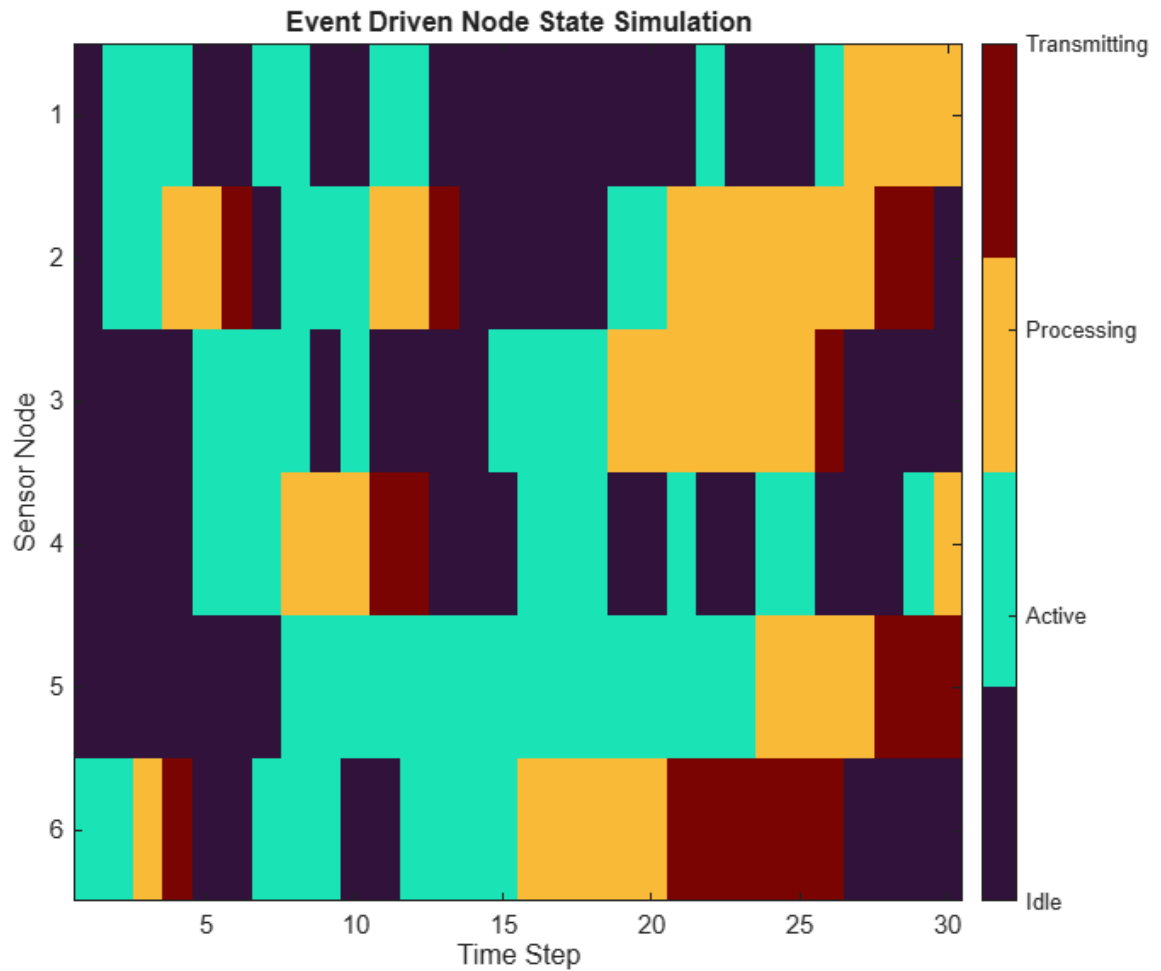
figure;
imagesc(nodeState);
colormap(turbo(4));
cb=colorbar;
cb.Ticks=1:4;
cb.TickLabels={'Idle', 'Active', 'Processing', 'Transmitting'};
xlabel('Time Step');
ylabel('Sensor Node');
title('Event Driven Node State Simulation');
set(gca, 'FontSize', 11);

figure;
for n=1:nodes
    subplot(nodes, 1, n);
    stairs(1:steps, nodeState(n, :), 'Linewidth', 1.6);
```

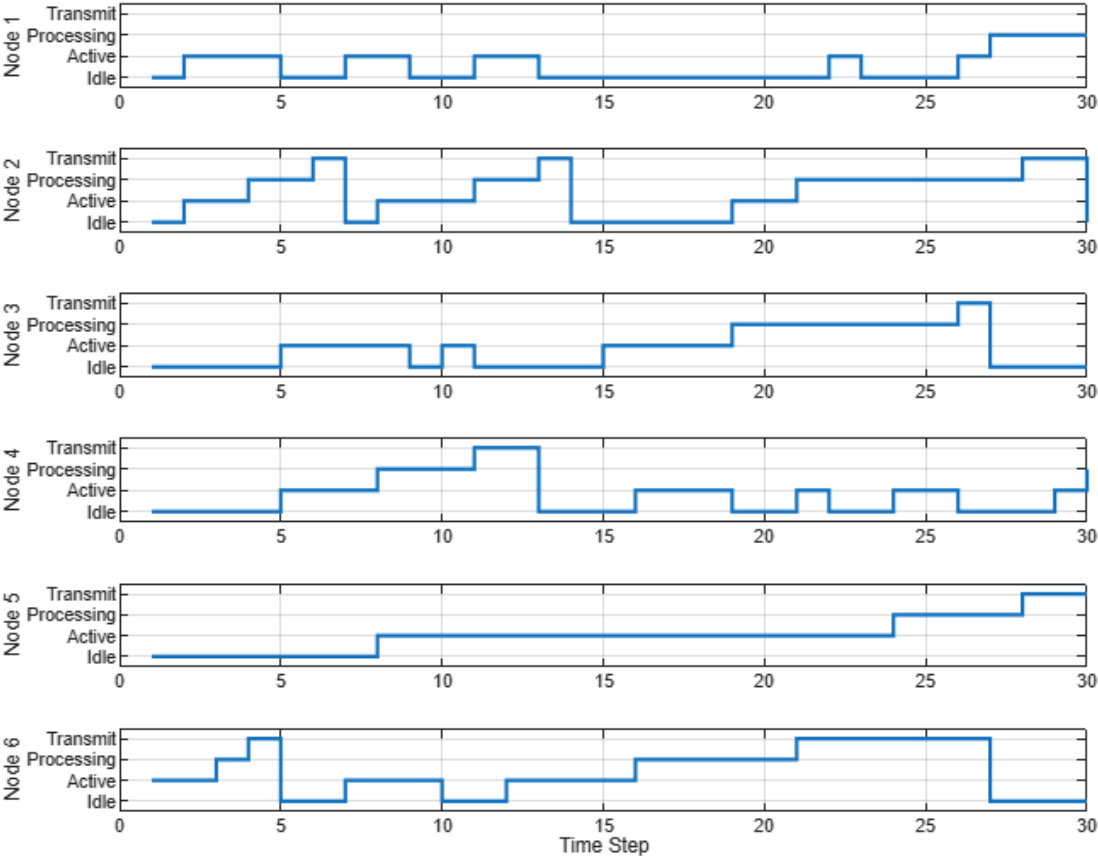
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ylim([0.5 4.5]);
yticks(1:4);
yticklabels({'Idle', 'Active', 'Processing', 'Transmit'});
ylabel(['Node ' num2str(n)]);
grid on;
end
xlabel('Time Step');
sgtitle('State Transitions of Multiple Nodes');

```



State Transitions of Multiple Nodes



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